# PROJECT REPORT

**Project Title**: SQL Data Analysis on an E-commerce Dataset

**Internship**: Decode Labs Internship – Project 3

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## 1. Introduction

For this project, I analysed an e-commerce dataset using PostgreSQL and pgAdmin 4. The goal was to apply SQL fundamentals to retrieve, organize and summarize data into meaningful business information.

The dataset contained sales transactions, customer information, products, payment methods and order statuses. Throughout the project, I wrote SQL queries to answer different business questions and understand the data better.

## 2. Project Objectives

The objectives of this project were to:

- Import a CSV dataset into PostgreSQL.
- Create a table that matches the dataset structure.
- Retrieve data using SQL.
- Filter records using conditions.
- Sort data in different orders.
- Group records to identify patterns.
- Perform calculations using SQL aggregate functions.
- Generate useful insights from the dataset.

## 3. Tools Used

- PostgreSQL
- pgAdmin 4
- CSV Dataset

## 4. Dataset Information

The dataset contains:

- 1,200 rows
- 14 columns

Some of the fields include:

- Order ID
- Customer ID
- Product
- Quantity
- Total Price
- Payment Method
- Order Status
- Referral Source
- Order Date

## 5. Methodology

**Step 1**: Creating the Database

I created a new PostgreSQL database that would store the e-commerce dataset.

This provided a dedicated environment for the project.

**Step 2**: Creating the Table

Before importing the CSV file, I created a table named ecommerce.

The table was created with the same column names and data types as the dataset. This was necessary because PostgreSQL imports data into an existing table.

Creating the table first ensured that every column in the CSV matched the correct column in the database.

**Step 3**: Importing the Dataset

After creating the table, I imported the CSV file into PostgreSQL using pgAdmin. The import was successful, and all 1,200 records were loaded into the table.

**Step 4**: Data Cleaning

During the analysis, I discovered that the totalprice column was stored as text because the values contained dollar signs ($) and commas.

**Example: $2,853.10**

Because SQL cannot calculate totals or averages from text values, the column needed to be converted into a numeric data type.

Instead of cleaning the values repeatedly in every query, I permanently converted the column using:

**ALTER TABLE ecommerce**

**ALTER COLUMN totalprice TYPE NUMERIC**

**USING REPLACE(REPLACE(totalprice,'$',''),',',''))::NUMERIC;**

This removed the dollar signs and commas before converting the values into numbers.

After this step, the column could be used directly with SQL functions such as **SUM()** and **AVG().**

## 6. SQL Analysis

The following analyses were carried out.

**Displaying Data**

- Displayed every record in the dataset.
- Displayed selected columns only.

Purpose:

To understand the dataset structure and retrieve only the required information.

**Filtering Records**

Used the WHERE clause to find:

- Delivered orders
- Laptop purchases
- Orders paid with Credit Card

Purpose:

To retrieve records that met specific conditions.

**Sorting Data**

Used ORDER BY to:

- Sort total price from highest to lowest.
- Sort quantity from lowest to highest.

Purpose:

To arrange records in a meaningful order.

**Counting Records**

Used COUNT() to determine:

- Total number of orders.
- Number of orders under each order status.

Purpose:

To understand transaction volume and compare order statuses.

**Calculating Total Sales**

Used SUM() to calculate the total revenue generated.

Purpose:

To determine overall sales.

**Sales by Payment Method**

Grouped transactions by payment method.

Purpose:

To identify which payment methods generated the highest sales.

**Sales by Referral Source**

Grouped transactions according to referral source.

Purpose:

To determine which marketing channels generated the highest revenue.

**Average Order Value**

Used AVG() to calculate the average value of customer orders.

Purpose:

To estimate the average amount spent per order.

**Average Order Value by Product**

Calculated the average sales value for each product.

Purpose:

To compare the average value generated by different products.

**Average Quantity Purchased**

Calculated the average quantity purchased for each product.

Purpose:

To understand purchasing behaviour.

**Customer Distribution**

Counted customers from each referral source.

Purpose:

To identify where customers came from.

## 7. Key Findings

Some of the findings from the analysis include:

- The dataset contains 1,200 customer orders.
- Total sales recorded amounted to 1,264,761.96.
- Different payment methods generated different levels of revenue.
- Referral sources contributed differently to sales performance.
- Products varied in both average order value and average quantity purchased.
- Orders were distributed across Delivered, Shipped, Pending, Returned and Cancelled statuses.

## 8. Challenges Encountered

One challenge encountered was that the totalprice column contained dollar signs and commas, preventing SQL from performing calculations.

Initially, the values were cleaned inside each query using the **REPLACE()** function.

A better solution was later implemented by permanently converting the column into the correct numeric data type using ALTER TABLE. This simplified the remaining queries and made the analysis cleaner.

## 9. Conclusion

This project improved my understanding of SQL fundamentals and showed how SQL can be used to answer practical business questions.

Working through the process of creating a table, importing data, cleaning a column, and performing analysis reinforced the importance of preparing data before generating insights.

Overall, the project provided hands-on experience with querying, filtering, grouping and summarizing data using PostgreSQL.